

HIV: POST-EXPOSURE PROPHYLAXIS (PEP)

Body fluids transmit infection through sexual contact (assault or consensual, burst condom), occupational exposure (sharps injury, splash to eye, mouth, nose or broken skin), sharing needles, contact with used condom and exposure to blood in sport or at accident scene. Tears, saliva (non-bloodstained), sweat, urine and stool are considered non-infectious fluids.

Give urgent attention to the patient exposed to infectious fluid:

- Exposure to blood, blood-stained fluid/tissue, pleural/pericardial/peritoneal/amniotic/synovial/cerebrospinal fluid, vaginal secretions, semen or breast milk
- Human bite severe enough to cause bleeding

Yes

Was there sexual contact, sharps injury, splash to eye/open wound/mouth/nose?

No

Yes

STEP 1. Give exposed patient immediate attention:

- If broken skin, clean area immediately with soap and water.
- If splash to eye, mouth or nose, immediately rinse mouth/nose or irrigate eye thoroughly with water.

No

STEP 2. Assess need for HIV PEP and give immediately if ≤ 72 hours since exposure:

Patient known HIV positive

Patient HIV negative or unknown: give first dose of HIV PEP (as below), obtain consent and do HIV rapid test ↗ 110.

Positive

Negative

One positive and one negative

Patient refuses HIV rapid test

Avoid giving HIV PEP, give instead routine HIV care ↗ 111.

Give HIV PEP only if ≤ 72 hours since exposure (ideally within 1 hour) for 28 days:

- Give **TDF** 300mg + **3TC** 300mg + **DTG** 50mg once daily for 28 days as a fixed combination tablet (TLD). If kidney disease, give instead **AZT** 300mg + **3TC** 150mg 12 hourly *plus* **DTG** 50mg once daily.
- Check for medication interactions and adjust doses if needed ↗ 118, especially TB and epilepsy treatment, contraceptives and other common medications like: calcium, iron, zinc, antacids, metformin.
- If source is failing 2nd line ART, discuss PEP with HIV hotline ↗ 178.

- If human bite severe enough to cause bleeding, prevent hepatitis B only. Risk of HIV transmission negligible and there is no need for HIV PEP.
- For all other exposures, reassure that HIV and hepatitis transmission is unlikely, avoid giving PEP.
- If unsure, discuss with HIV hotline ↗ 178 or specialist.

STEP 3. Take blood from exposed patient: if giving TDF, do creatinine (eGFR). If giving AZT, do FBC + differential count.

Send exposed patient's blood for HBsAb titre (unless occupational exposure with documented HBsAb titre ≥ 10).

STEP 4. Take blood from source, if possible and source consents:

- Do rapid HIV test ↗ 110 and send blood for HBsAg and hepatitis C antibody. If case of human bite, only do HBsAg. If sexual exposure, also do syphilis.
- If source HIV rapid negative, no need to continue PEP unless signs of acute HIV infection (history of flu-like symptoms - fever, sore throat, headache). If unsure, discuss PEP with HIV hotline ↗ 178.

STEP 5. Give hepatitis B PEP if needed:

If exposed patient has not previously had 3 doses of hepatitis B vaccine or unsure, give 1st dose of **hepatitis B vaccine** 1mL IM. If sharps injury, give within 7 days. If sexual exposure, within 14 days.

STEP 6. Continue to assess and manage the patient:

If sexual assault ↗ 88. If emergency contraception needed ↗ 154. Review patient and blood results within 3 days → 109.

AZT – zidovudine; **DTG** – dolutegravir; **FTC** – emtricitabine; **TDF** – tenofovir; **3TC** – lamivudine; **HBsAg** – Hepatitis B surface antigen; **HBsAbs** – Hepatitis B surface antibody